NeuroImage 44 (2009) 200–212

Contents lists available at ScienceDirect

NeuroImage

journal homepage: www.elsevier.com/locate/ynimg

Dissociable neural mechanisms for determining the perceived heaviness of objects and the predicted weight of objects during lifting: An fMRI investigation of the size–weight illusion

Philippe A. Chouinard [a,*], Mary-Ellen Large [a,b], Erik C. Chang [a,c], Melvyn A. Goodale [a]

[a] CIHR Group on Action and Perception, Department of Psychology, University of Western Ontario, 1151 Richmond Street, London, Ontario, Canada, N6A 5B8
[b] Department of Psychology, University of Hull, Hull, UK
[c] Institute of Cognitive Neuroscience, National Central University, Jhongli, Taoyuan, Taiwan

ARTICLE INFO

Article history:
Received 29 May 2008
Revised 8 August 2008
Accepted 9 August 2008
Available online 28 August 2008

Keywords:
Size–weight illusion
Object lifting
Weight perception
Functional magnetic-resonance imaging
Ventral premotor area
Sensorimotor areas

ABSTRACT

In size–weight (SW) illusions, people learn to scale their fingertip forces for lifting small and big objects of equal weight even though they fail to learn perceptually that both objects have the same weight. The question then arises as to what the separate neural mechanisms are for determining the perceived heaviness of objects and the predicted weight of these objects during lifting. To answer this question, we used fMRI to first identify areas that code for the size, weight, and density of objects using an adaptation paradigm. We then contrasted BOLD in the SW illusion condition in which subjects falsely perceived the smaller of two equally weighted objects as heavier versus a condition in which size and weight did not differ between objects. Sensory areas in the parietal and temporal cortex adapted to the size of objects and the primary motor area (M1) contralateral to the lifting hand adapted to the weight of objects. The ventral premotor area (PMv), which did not adapt to either the size or the weight of objects, adapted instead to the density of objects, and responded more when subjects falsely perceived differences in weight between objects in the SW illusion condition. Taken together, we conclude that the real-world properties of objects, such as size and weight, are computed by sensory areas and by M1 respectively, whereas the perceived heaviness of objects, presumably based on their apparent density, is computed by PMv, a higher-order area well placed to integrate sensory information about the size of objects and the weight of objects.

© 2008 Elsevier Inc. All rights reserved.

Introduction

In size–weight (SW) illusions, people perceive the smaller of two equally weighted objects as heavier when those objects are lifted in succession. Recent behavioral studies reveal that SW illusions occur independently of sensorimotor predictions (Flanagan and Beltzner, 2000; Grandy and Westwood, 2006; Chang et al., 2008). In these studies, people quickly learned to scale their fingertip forces to the actual weight of objects, even though they failed to learn perceptually that both objects had exactly the same weight. In other words, SW illusions persisted despite the fact that fingertip forces were exquisitely matched to the weight of the objects. These studies reveal therefore that SW illusions are driven by perceptual processes that operate independently of the sensorimotor system. Although SW illusions have been studied extensively for over a century (Charpentier, 1891), little is known about the neural mechanisms responsible for these illusions. Nobody to our knowledge has used functional neuroimaging to examine what brain areas might be involved in generating SW illusions.

One way to address this issue is to use functional magnetic-resonance imaging (fMRI) to first identify areas that code for the size, weight, and density of objects using an adaptation paradigm and then contrast blood oxygenation-level dependent signal (BOLD) in a condition in which subjects falsely perceive the smaller of two equally weighted objects as heavier during the SW illusion versus a condition in which size and weight do not differ between objects. We hypothesized that an area involved in driving SW illusions would most likely be an area that also computes the density of objects. A number of studies have demonstrated that changes in density, independent of changes in mass, can influence people's judgments of the heaviness of objects (Grandy and Westwood, 2006; Kawai, 2002; Ross 1969). According to this line of research, people make decisions about the weight of objects based on the feature that is most salient for understanding their heaviness—in the case of SW illusions, the respective density of the two objects. One might expect therefore that a brain area involved in generating SW illusions would have to be able to compute density by integrating sensory information about the size of objects and the weight of those objects when they are lifted. Indeed, we would predict that such an area would not adapt to the repeated presentation of objects of the same size or weight, but would adapt instead to the repeated presentation of objects that have the same density.

* Corresponding author.
E-mail address: pchouin@uwo.ca (P.A. Chouinard).

1053-8119/$ – see front matter © 2008 Elsevier Inc. All rights reserved.
doi:10.1016/j.neuroimage.2008.08.023

Repetition suppression effects have been studied at the level of both cortical neurons in monkeys and local hemodynamic changes in the brain of both monkeys and humans (for review, see Grill-Spector et al., 2006). In the case of neuroimaging, if a given area contains neurons that code for a particular object feature, then the BOLD response would be expected to be higher during conditions in which that feature changes across trials as compared to the BOLD response during conditions in which that feature remains the same. When other features of objects that are not coded by neurons in this region are manipulated, there should be no difference in the BOLD response between conditions in which that feature if varied or remains the same. This technique is sometimes called fMRI adaptation. Because fMRI adaptation has proved to be an effective way of determining what is and what is not processed by a particular brain area, it is being used increasingly by the neuroimaging community in both vision (Grill-Spector et al., 2006) and motor (Dinstein et al., 2007) research.

We used a 2×2 factorial design in which subjects lifted pairs of objects that varied or did not vary in size and that varied or did not vary in weight (Fig. 1). As a consequence of manipulating both size and weight in this manner, we also created conditions in which the density of pairs of objects either varied or remained the same. Thus, the manner in which the conditions were set-up enabled us to examine the effects of size, weight, and density independently from one another (for further explanation, see Fig. 1). In other words, the design allowed us to identify brain areas that adapted to size (main effect of size), brain areas that adapted to weight (main effect of weight), and brain areas that adapted to density (size×weight interaction). We tested also for differences in BOLD when subjects falsely perceived the smaller of two equally weighted objects as heavier in the SW illusion condition versus a condition in which size and weight did not differ between pairs of objects.

Our imaging data revealed that sensory areas in the parietal and temporal cortex adapted to the size of objects whereas the primary motor area (M1) contralateral to the lifting hand adapted to the weight of objects. The left ventral premotor area (PMv), which did not adapt to either the size or the weight of objects, adapted instead to the density of objects, and responded more when subjects falsely perceived differences in weight between objects in the SW illusion condition. Based on these results, we conclude that the real-world properties of objects, such as size and weight, are computed by sensory areas in the parietal and temporal cortex and by M1 respectively, whereas the perceived heaviness of objects, presumably based on their apparent density, is computed by PMv, a higher-order area well placed to integrate sensory information about the size of objects and the weight of those objects after they have been lifted.

Methods

Twelve subjects (8 females, age range 18–31 years, mean=21.7 years) participated in a behavioral experiment and twelve different subjects (5 females, age range 20–34 years, mean=26.3 years) participated in

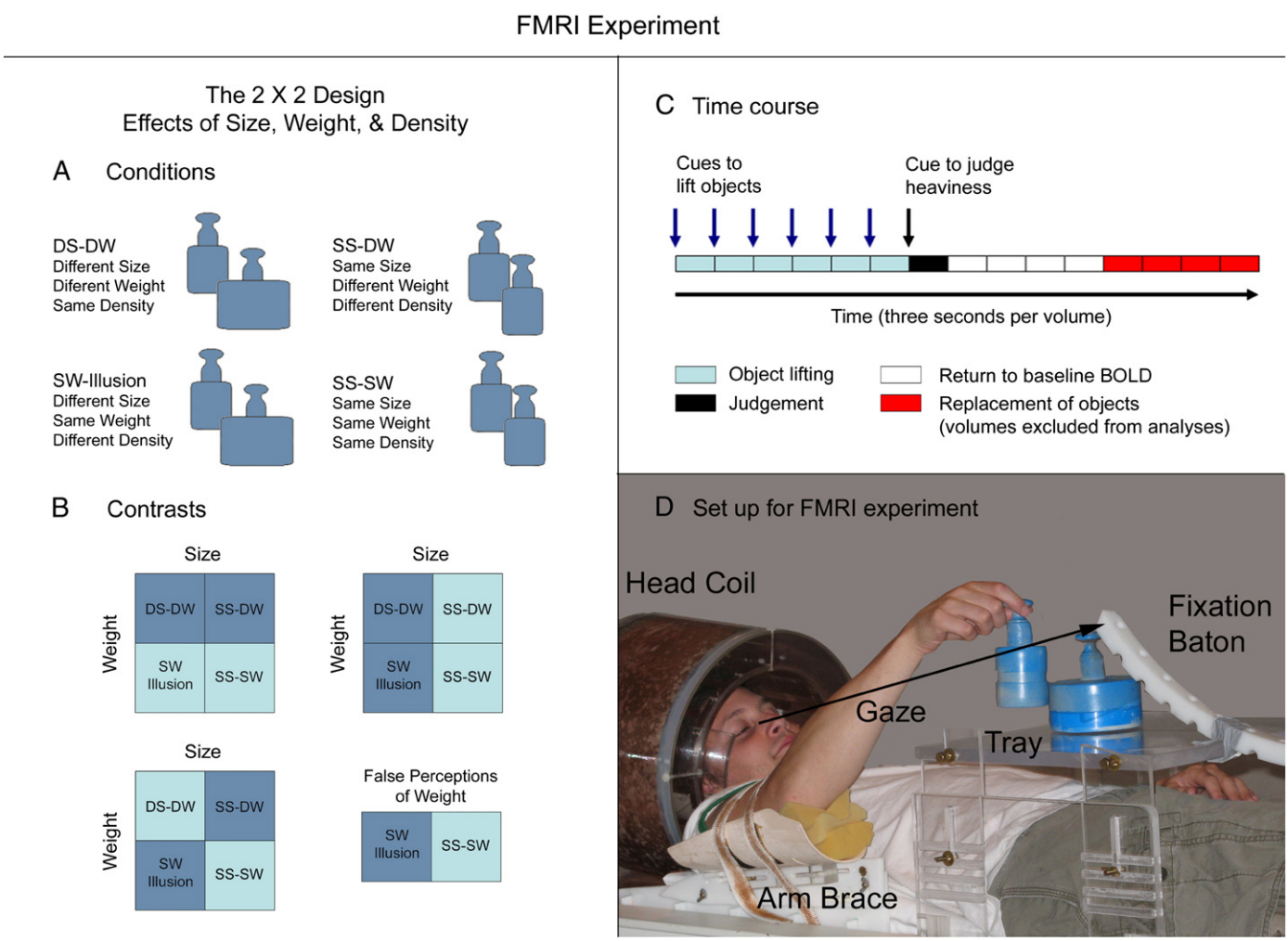

Fig. 1. fMRI experiment. The figure provides illustrations of the conditions (A), the contrasts (B), the time course (C), and the set-up (D) used in the fMRI experiment. See "Overview" in Methods under "fMRI experiment" for further details about this figure.

an fMRI experiment. All subjects had a right-hand preference as determined by a questionnaire (Oldfield, 1971) and provided informed written consent. Subjects reported that they were naïve about the SW illusion, had corrected-to-normal visual acuity, and no history of neurological impairments. The Research Ethics Board of the University of Western Ontario (London, Ontario, Canada) approved all procedures.

Subjects in the fMRI experiment underwent a training session prior to scanning to familiarize them with the procedures and to allow them to practice lifting objects with as little motion as possible in the shoulder (so that any conduction of arm movements to the head during fMRI acquisition would be diminished). For statistical analyses of the fMRI data, we first identified key brain areas involved in lifting objects by contrasting all conditions pooled versus rest and then directed all subsequent analyses in each of these defined 'object-lifting' regions-of-interest (ROI) using both (1) a directed approach for group voxel-wise analyses (i.e. reducing the criterion for reporting a peak in a ROI as significant; Worsley et al., 1996), and (2) a subject-by-subject ROI approach for analyzing imaging data. This was done so that we could examine the more subtle effects of fMRI adaptation to the size, weight, or density of objects in key areas brain areas involved in lifting objects.

Task objects

To make objects, we cut plastic pipes to a desired length and placed plastic covers on either side (Figs. 1A, D). Wooden handles of identical dimensions were attached to the top of each object. Subjects manipulated all objects using a similar grasping configuration. We adjusted the weight of each object by inserting lead pellets in the objects and placed foam around the pellets so that the center of mass of the objects matched their geometric centers. Small objects had a radius of 3 cm and a height of 7 cm and weighed either 136 g (density = 0.69 g/cm^3) or 500 g (density = 2.53 g/cm^3). Big objects had a radius of 5.75 cm and a height of 7 cm and weighed either 136 g (density = 0.19 g/cm^3) or 500 g (density = 0.69 g/cm^3). In the different size–different weight (DS–DW) condition, we also included a small object with a radius of 3 cm and a height of 10.3 cm that weighed 200 g (density = 0.69 g/cm^3) and a big object with a radius of 5.75 cm and a height of 8.4 cm that weighed 600 g (density = 0.69 g/cm^3). The addition of two more objects in the DS–DW condition was done to balance the number of different types of objects presented in each condition. In the behavioral experiment, handles equipped with force sensors mounted on objects resulted in slightly heavier objects (+60 g).

Behavioral experiment

Overview

A behavioral experiment was carried out first to ensure that the task objects that we planned to use in the fMRI experiment could generate powerful SW illusions. In addition, we felt it was important to replicate previous work showing that people can learn quickly to scale their fingertip forces correctly in the context of SW illusions (Flanagan and Beltzner, 2000; Grandy and Westwood, 2006; Chang et al., 2008). In other words, we wanted to verify that subjects could generate appropriate sensorimotor predictions even though they continue to perceive the smaller object as heavier than the larger one. The experiment had three conditions: two SW illusion conditions and a DS–DW control condition (Fig. 2). In the two SW illusion conditions, subjects lifted two different-sized objects with their right hand that were both either 560 g or 200 g. In the DS–DW condition, subjects lifted a big object weighing 560 g and a small object of equal density

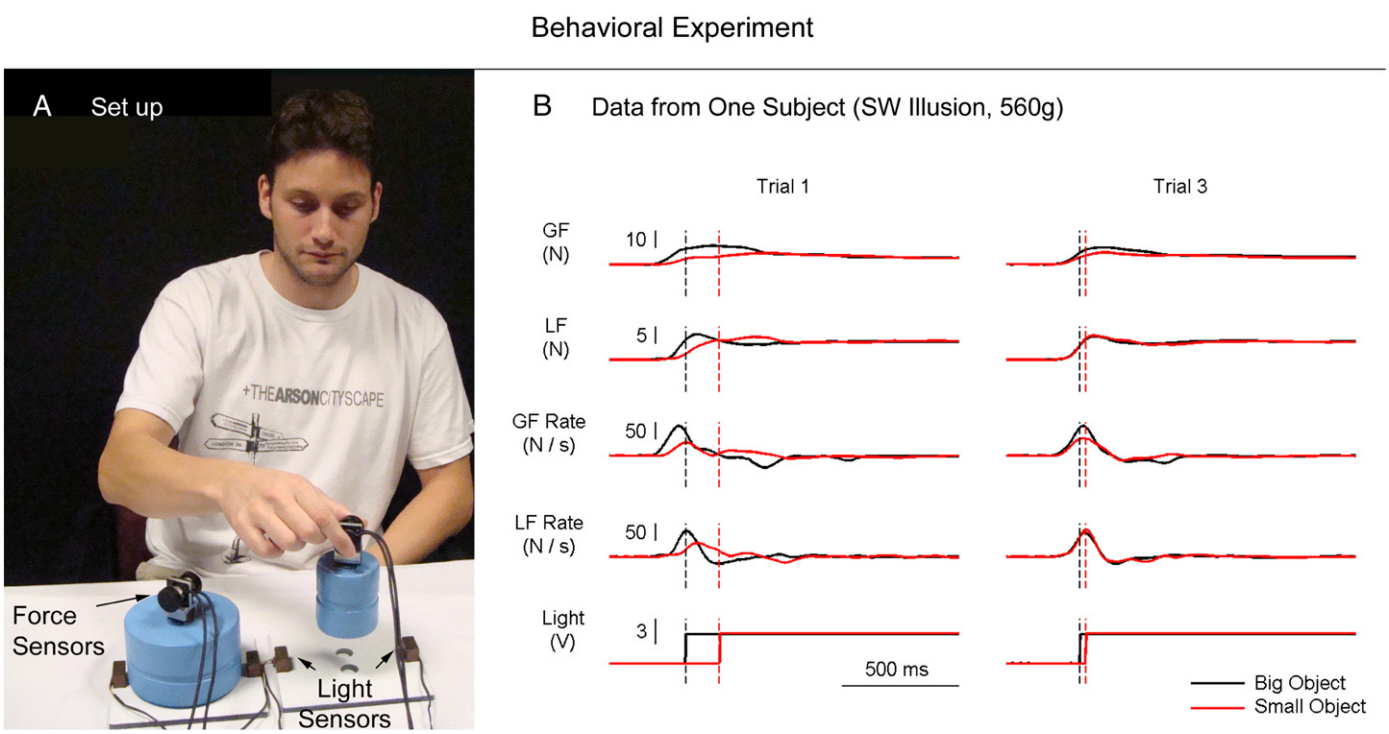

Fig. 2. Behavioral experiment. (A) In the behavioral experiment, subjects lifted two different-sized objects with their right hand fifteen times that were both either 560 g or 200 g (the two SW illusion conditions) or 560 g for the big object and 200 g for the small object (different size–different weight, DS–DW condition). Objects in the DS–DW condition had the same density (see Task objects in Methods). Force sensors mounted on the objects recorded grip and load forces applied during lifting and light sensors were used to detect lift-off. (B) The panel shows signals for grip force (GF, row 1), load force (LF, row 2), rates in grip force (row 3), rates in load force (row 4), and light sensors for detecting lift-off (row 5) in one subject lifting big (black) and small (red) objects of 560 g. Subjects all applied increasing amounts of force until sufficient load force was reached to overcome gravity on the object. When the small object was lifted in trial 1, the initial increase in load force was usually insufficient to cause lift-off and load forces had to increase again before lift-off (red-dotted line) could occur. When the big object was lifted in trial 1, all subjects initially applied too much force causing lift-off (black-dotted line) to occur earlier than expected. By trial 3, participants were better at scaling forces for actual weight, and hence, differences when lift-off (dotted lines) occurred were diminished between the two objects. Signals are aligned at the time participants initiated load forces greater than 0.2 N.

weighing 200 g for the small object. Each object was lifted fifteen times. Given that each block in the fMRI experiment had a total of three trials for each object (see below in fMRI Overview), we compared perceptual estimates of weight as well as grip and load forces on trials 1, 3, and 15.

Collection and analyses of the data

Each object was equipped with two six-axis force–torque sensors (Nano17 F/T; ATI Industrial Automation, Garner, North Carolina) that measured forces applied by the fingers in three dimensions (Fig. 2A). Subjects lifted each object by the tips of their thumb and index finger of their right hand fifteen times for 3 s after the onset of auditory cues. The order of lifts was randomly generated in such a way that subjects never lifted the same object more than twice in a row. We counter-balanced across subjects the order of conditions (i.e. SW illusion condition with 560 g objects, SW illusion condition with 200 g objects, and the DS–DW condition). For perceptual estimates of weight, we used a magnitude estimation procedure (described in more detail by Zwislocki and Goodman, 1980). In brief, subjects rated the heaviness of the object by reporting an integer; the heavier the object, the larger the number. No constraints on the range of this integer were provided, and participants were allowed to use a range they found comfortable. Estimates were transformed as a percent of a subject's maximum estimate during the experiment. Signals from the force and light sensors were sampled at 400 Hz. Load force was defined as the resultant force tangential to the surface of the handle and grip force normal to the surface of the handle. Force signals were smoothed using a fourth-order, zero-phase lag, low-pass Butterworth filter with a cutoff frequency of 14 Hz, and were differentiated using a three-point central difference equation to calculate rates in force. For simplicity, we report grip forces only for the index finger (the thumb yielded similar results given that participants lifted objects vertically). For statistical analyses, we performed ANOVA with object and trial number as within-subject factors. Only trials 1, 3 (total number of trials in the fMRI experiment), and 15 were included in the ANOVA. Tukey's HSD tests, which corrected for multiple comparisons, were used to make pair-comparisons.

fMRI experiment

Overview

Fig. 1 provides an overview of the fMRI experiment. We used a 2 × 2 factorial design in which subjects lifted pairs of objects that were different or similar in size and different or similar in weight—while ensuring the density between pairs of objects was the same when subjects lifted objects that were different in both size and weight.

Fig. 1A illustrates the four conditions that we used in the experiment. Subjects lifted pairs of objects that were either different or similar in size and either different or similar in weight. Pairs of objects in the DS–DW condition had different sizes and different weights while density was always the same between the two objects. Pairs of objects in the SW illusion condition had different sizes and had the same weight (density therefore differed between the two objects). Pairs of objects in the SS–DW condition had different weights and had the same size (density therefore also differed between the two objects in this condition). Pairs of objects in the SS–SW condition were always identical (see Task objects in Methods for further details concerning the physical properties of the objects and see Table 1 for further details concerning the frequencies with each pair of objects were presented).

Fig. 1B illustrates the contrasts that we used. To examine effects of size, we compared BOLD in which objects differed in size versus objects of the same size (this removed effects of "real" weight and density). To examine effects of "real" weight, we compared BOLD in which objects differed in weight versus objects of the same weight (this removed effects of size and density). To examine effects of density, we compared BOLD in which objects differed in density versus objects of the same density (this removed effects of size and "real" weight). Note that this contrast corresponds also to the interaction effect between size and weight. Any effect of weight so far pertains to "real" weight and not "perceived" weight. Therefore, we also tested for differences in BOLD between the SW illusion and SS–SW conditions — which we will refer to as the "illusory" contrast. To determine whether or not differences in BOLD in the illusory contrast were driven by differences in density or by differences in size (or both) between the two objects in the SW illusion condition, we performed two conjunction analyses (Price and Friston, 1997). In the first, we

Table 1
Weight, size, and density of task objects (fMRI experiment)

DS–DW		SW illusion		SS–DW		SS–SW	
Object pairs presented	Frequency presented	Object pairs presented	Frequency presented	Object pairs presented	Frequency presented	Object pairs presented	Frequency presented
Weight of objects (in g)							
136 and 500	50%	136 and 136	50%	136 and 500	50%	136 and 136	50%*
200 and 600	50%	500 and 500	50%	136 and 500	50%	500 and 500	50%*
Average weight Lifted = 350		Average weight Lifted = 318		Average weight Lifted = 318		Average weight Lifted = 318	
Size of objects (in cm^3)							
198 and 727	50%	198 and 727	50%	198 and 198	50%	198 and 198	50%†
291 and 872	50%	198 and 727	50%	727 and 727	50%	727 and 727	50%†
Average size Lifted = 522		Average size Lifted = 231		Average size Lifted = 231		Average size Lifted = 231	
Density of objects (in g/cm^3)							
0.69 and 0.69	100%	0.69 and 0.19	50%	0.69 and 2.53	50%	0.69 and 0.69	50%
		0.69 and 2.53	50%	0.69 and 0.19	50%	0.19 and 0.19	25%
						2.53 and 2.53	25%
Average density Lifted = 0.69		Average density Lifted = 1.02		Average density Lifted = 1.02		Average density Lifted = 1.02	

The table shows which pairs of objects were used in each condition (with their frequency of presentation), as well as the average size, weight, and density of objects that were lifted in each condition during the fMRI experiment. In the SS–SW condition, frequencies presented for object pairs both weighing 136 g can be divided further into 25% for smaller objects and 25% for bigger objects and frequencies presented for object pairs both weighing 500 g can be divided further into 25% for smaller objects and 25% for bigger objects (*). Likewise, frequencies presented for object pairs that were both 198 cm3 in size can be divided further into 25% for lighter objects and 25% for heavier objects and frequencies presented for object pairs that were both 727 cm3 in size can be divided further into 25% for lighter objects and 25% for heavier objects (†).

conjoined the illusory contrast with the contrast that identified voxels that showed effects of density and in the second, we conjoined the illusory contrast with the contrast that identified voxels that showed effects of size. We offer a more detailed discussion on the logic underlying each of these contrasts in Supplementary material.

The fMRI experiment had a block design (Fig. 1C). Each block consisted of 6 lifts. Subjects lifted each object by the tips of their thumb and index finger of their right hand after hearing auditory cues that indicated which of the two objects they had to lift. Each lift lasted 3 s. Each object was lifted three times in a block. The order of lifts was randomly generated in such a way that subjects never lifted the same object more than twice in a row. Subjects then heard a different cue instructing them to either wiggle their left foot if they thought the object on their left was heavier or their right foot if they thought the object on their right was heavier. Expected responses by either the left or right foot were balanced across conditions (and no differences were found among conditions in the percentage of times participants actually responded with a particular foot). Twelve seconds ensued to allow the hemodynamic response to return to baseline levels and the experimenters then replaced the objects. We counterbalanced the order of conditions in a given subject as well as the relative positioning of the objects on the tray. A total of eight runs were collected in each participant. In a given run, participants completed two blocks for the DS–DW, SW Illusion, and SS–DW conditions and four blocks for the SS–SW condition. Extra blocks of the SS–SW condition were required to ensure that all four types of objects were presented. This yielded a total of 16 blocks each for the DS–DW, SW Illusion, and SS–DW conditions and a total of 32 blocks for the SS–SW condition.

Fig. 1D illustrates how we positioned participants inside the scanner. We positioned subjects with their heads tilted to enable them to look at the objects placed on a tray over their legs. Subjects lay supine with the torso tilted by a shallow ramp. Within the cylindrical head coil, the head was tilted approximately 30° with foam wedges and objects fell within the lower visual-field, a configuration that is not unusual in everyday grasping. Subjects were instructed to maintain fixation on a plastic baton throughout a given run. An arm brace supported and restrained their upper right arm. This allowed limited motion at the elbow and full motion at the wrist while restricting shoulder movement and the conduction of arm movements to the head.

fMRI acquisition

Scanning was performed on a 4-T Varian/Siemens MRI system (Erlangen, Germany). An anatomical scan was performed encompassing the whole brain after half of the functional runs were completed. This was achieved by collecting 96 two-mm thick slices using a 3-D T1-weighted magnetization prepared FLASH imaging sequence (TI = 500 ms, TE = 5.5 ms, TR = 10.0 ms, flip angles = 11 and 180°). The in-plane resolution of the anatomical scans was 256 pixels × 256 pixels. To collect functional data, we used a T2*-weighted four-shot echo-planar imaging sequence (TE = 15.0 ms, TR = 750 ms, flip angle = 30°) for BOLD acquisition (Ogawa et al., 1992). Acquisition time was 3 s per volume. The field of view was 19.2 cm × 19.2 cm with an in-plane matrix size of 64 pixels × 64 pixels. Each image covered the whole brain except for the temporal poles and consisted of 25 five-mm thick slices oriented parallel to a line connecting the base of the cerebellum to the base of the orbitofrontal cortex (voxel size = 3.00 × 3.00 × 5.00 mm). There were no gaps between slices and 152 volumes were collected in a run.

Preprocessing of fMRI data

We used AFNI (National Institutes of Health, Bethesda, MD; see Cox, 1996; available at http://afni.nimh.nih.gov/afni) for motion correction. This aligned all functional data to the fifth volume of the run closest to the anatomical scan. Inspection of the data revealed that head movements associated with lifting objects, when present, were within an acceptable range of less than 1 mm (see Supplementary material). We transformed each subject's anatomical MRI into standardized space using an automated feature-matching algorithm and the average-305 brain from the Montreal Neurological Institute (MNI) as a template (see Collins et al., 1994; both available at http://www.bic.mni.mcgill.ca/software/distribution), and then aligned each subject's functional data to their anatomical MRI in standardized space. For group voxel-wise analyses, we spatially smoothed the data with a 6-mm full-width half-maximum Gaussian filter to reduce effects of inter-subject variability in brain anatomy. For subject-by-subject analyses, we did not smooth the data so as to not diminish spatial sensitivity. We then imported the functional data into FMRISTAT (Montreal Neurological Institute, Montreal, QC, Canada; see Worsley et al., 2002; available at http://www.math.mcgill.ca/keith/fmristat/) to perform voxel-wise analysis based on a linear model with correlated errors. We excluded from the analyses the first four volumes in each run, as these volumes may not reflect steady state, and every four volumes when the experimenters changed objects due to possible perturbations in the magnetic field. We removed drift by adding third-order polynomial covariates in the volume acquisition times in the design matrix and expressed the signal value at each voxel as a percentage of the signal averaged across the whole volume. The time course of the response was modeled for each condition convolved with a hemodynamic-response function (Glover, 1999). A predictor of no interest, corresponding to when subjects reported which object they thought weighed heaviest, was also included in the model. This removed the effects of foot responses from the BOLD analyses.

Group voxel-wise analyses

All group voxel-wise analyses were conducted using a random-effects model. We examined effects of size, weight, and density (Figs. 1A, B) by testing at a given voxel for differences in BOLD when subjects lifted objects of different sizes compared to objects of the same size, objects of different weights compared to objects of the same weight, and objects of different densities compared to objects of the same density. We also tested at a given voxel for differences in BOLD when subjects falsely perceived differences in weight during the SW illusion versus the SS–SW condition (i.e. the "illusory" contrast, Figs. 1A, B). To determine whether or not differences in BOLD in our illusory contrast were driven by differences in density or by differences in size (or both) between two objects in the SW illusion condition, we conjoined the illusory contrast with the contrast identifying voxels that showed effects of density and conjoined the illusory contrast with the contrast identifying voxels that showed effects of size.

We used the minimum given by Bonferroni correction and random-field theory (Worsley et al., 1992) to correct for multiple comparisons for exploratory searches of the whole brain and directed searches in key brain areas involved in lifting objects. For exploratory searches, this yielded a threshold of $T_{(11)} = 8.0$ for reporting a peak as significant at $P_{corr} < 0.05$ in an estimated grey-matter volume of 700 cm^3. For directed searches in key brain areas involved in lifting objects, a mask of 'object-lifting' ROIs was first created to encompass all voxels with a p-uncorrected value of less than 0.001 in each brain area that was found to be significantly activated during the lifting of objects (i.e. all conditions pooled versus rest). We then superimposed our statistical maps, masked for the 'object-lifting' ROIs, over the average anatomical MRI of all subjects in standardized space and report coordinates of significant peaks. Values equal to or exceeding $T_{(11)} = 4.0$ were considered significant at $P_{corr} < 0.05$ ($P_{uncorr} < 0.001$) in search volumes of 2.5 cm^3. For conjunction analyses, we thresholded the statistical maps for directed searches at $P_{uncorr} < 0.005$. In other words, a given voxel in an 'object-lifting' ROI had to therefore yield a value of $P_{uncorr} < 0.005$ in each of the two conjoined contrasts to be considered significant. This approach for conjunction analyses is described in more detail by Nichols et al. (2005).

Subject-by-subject analyses

Given that some of the 'object-lifting' ROIs were in close proximity to each other and the problem of inter-subject variability in brain anatomy in standardized space (Brett et al., 2002), it was deemed necessary to also perform subject-by-subject analyses to disambiguate effects occurring in separate 'object-lifting' ROIs that were in close proximity to each other. Each individual's statistical map from all conditions pooled versus rest was first superimposed over their anatomical MRI. We then created 'object-lifting' ROIs by drawing spheres with a radius of 3 mm centered on the 'hotspot' of activity for each region found to be significantly activated during the lifting of objects (see Regions activated during object lifting in Results) provided that activity in each individual reached $P_{uncorr}<0.001$ and fell within appropriate landmarks. ROIs for M1 were in the 'hand knob' of the precentral gyrus (Yousry et al., 1997). ROIs for S1 were in the postcentral gyrus posterior to the 'hand knob' of the precentral gyrus. ROIs for AIP were in the postcentral sulcus close to where this sulcus joins the intraparietal sulcus (Frey et al., 2005). ROIs for SPL were in the anterior part of the superior-parietal lobule. ROIs for PMv were in the ventral part of the precentral sulcus (Chouinard and Paus, 2006). ROIs for the thalamus were in the ventro-lateral portion of the thalamus. ROIs for Fus were in the fusiform gyrus. ROIs for the anterior lobule of the cerebellum were above and anterior to the primary fissure of the cerebellum. ROIs for the posterior lobule of the cerebellum were below and posterior to the horizontal fissure of the cerebellum. Using these functional and anatomical criteria, we were able to formulate each of these ROIs in all subjects. After formulating 'object-lifting' ROIs in each subject, we extracted their percent-signal change in BOLD for each condition. ANOVA was performed on these extracted values to examine effects of size, weight, and density with size and weight as within-subject factors. Tukey's HSD tests, which corrected for multiple comparisons, were used to make pair-wise comparisons.

Results

Behavioral experiment

Prior to a set of trials, when asked to visually examine a pair of objects, subjects unanimously reported that they expected the bigger object to be heavier. After completing a set of trials, subjects

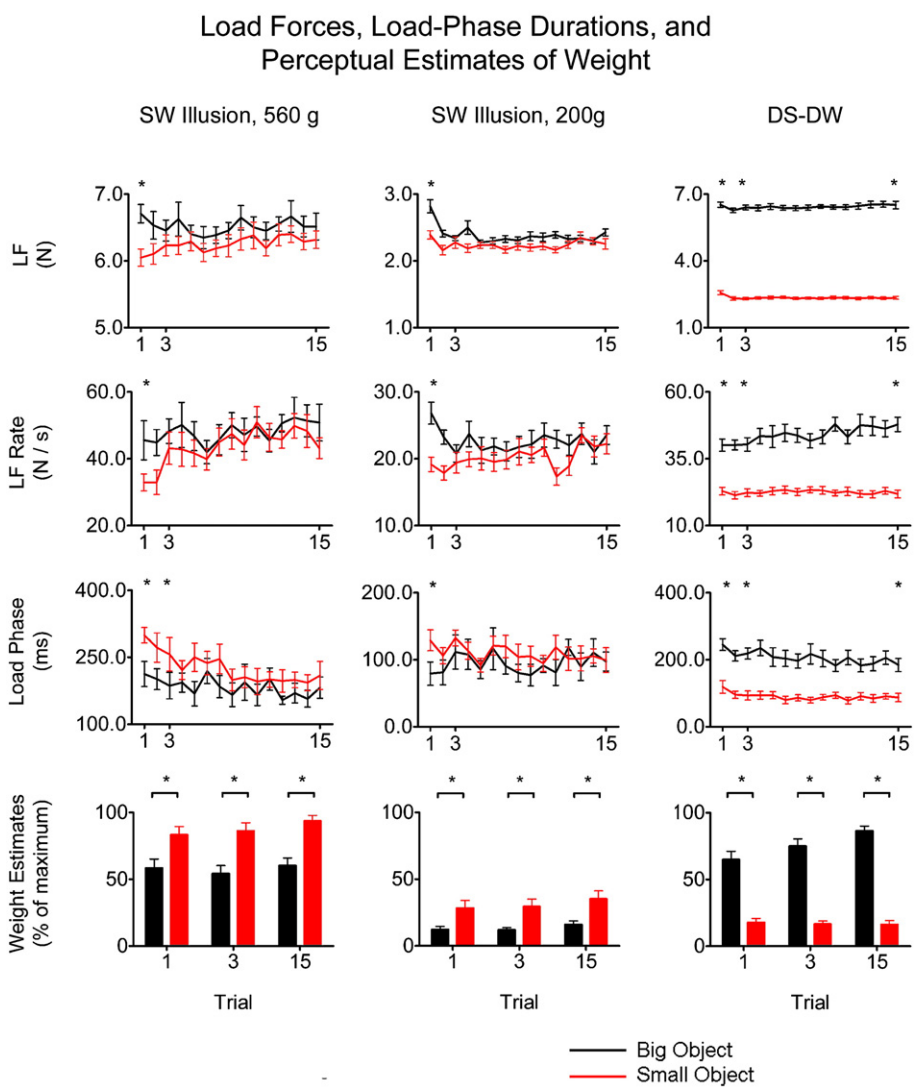

Fig. 3. Load forces, load-phase durations, and perceptual estimates of weight. The graphs plot means ± SEM for peak load forces (LF, row 1), peak rates in load force (row 2), load-phase durations (row 3), and perceptual estimates of weight as a percentage of each subject's maximum estimate (row 4) measured in the behavioral experiment. The plots illustrate measurements when participants lifted big (black) and small (red) objects that weighed either both 560 g (column 1) both 200 g (column 2), or 560 g for the big object and 200 g for the small object (column 3). We present grip forces in Supplementary material. *Y*-axes are scaled differently to better distinguish differences in measurements obtained between objects. Asterisks (*) denote significant differences between big and small objects. Given that the total number of trials in a block for the fMRI experiment was three, we compared measurements obtained between objects on trials 1, 3, and 15.

unanimously reported that the smaller object was heavier in SW illusion conditions, and thus experienced SW illusions. Subjects also reported that the bigger object was heavier in the DS–DW condition. Fig. 2B provides force signals acquired in one subject to illustrate how subjects typically lifted two-equally weighted objects of different sizes. As shown, subjects applied increasing amounts of force until sufficient load force was reached to overcome gravity on the object. When the small object (red-solid line) was lifted in trial 1, this initial increase in load force was usually insufficient to cause lift-off and load forces had to increase again before lift-off could occur (red-dotted line). When the big object (black-solid line) was lifted in trial 1, subjects initially applied too much force causing lift-off to occur prematurely (black-dotted line). By trial 3, however, subjects were better at scaling forces for actual weight, and hence, differences when lift-off (dotted lines) occurred were diminished between the two objects.

Fig. 3 illustrates peak load forces, peak rates in load force, load-phase durations, and perceptual estimates of weight. We present peak grip forces and peak rates in grip force in Supplementary material given that it is the load forces that overcome the effects of gravity on objects and thus it is these forces that are most closely linked with expectations of weight (Johansson and Westling, 1988). In SW illusion conditions, perceptual estimates of weight were all greater for the smaller object after completion of trials 1, 3, and 15 (all $P<0.005$). Peak forces and peak rates in force were all greater when lifting the bigger object on trial 1 (all $P<0.05$). Shorter load-phase durations were also observed for the bigger object (both $P<0.05$). By trial 3, load forces no longer differed between the two objects (Fig. 3), but grip forces did still differ in some cases (see Supplementary material). None of the force measurements differed between the two objects on trial 15. Conversely, perceptual estimates of weight in the DS–DW condition were all greater for the bigger object after completion of trials 1, 3, and 15 (all $P<0.0001$). Peak forces, peak rates in force, and load-phase durations were all greater when lifting the bigger object on trials 1, 3, and 15 (all $P<0.05$).

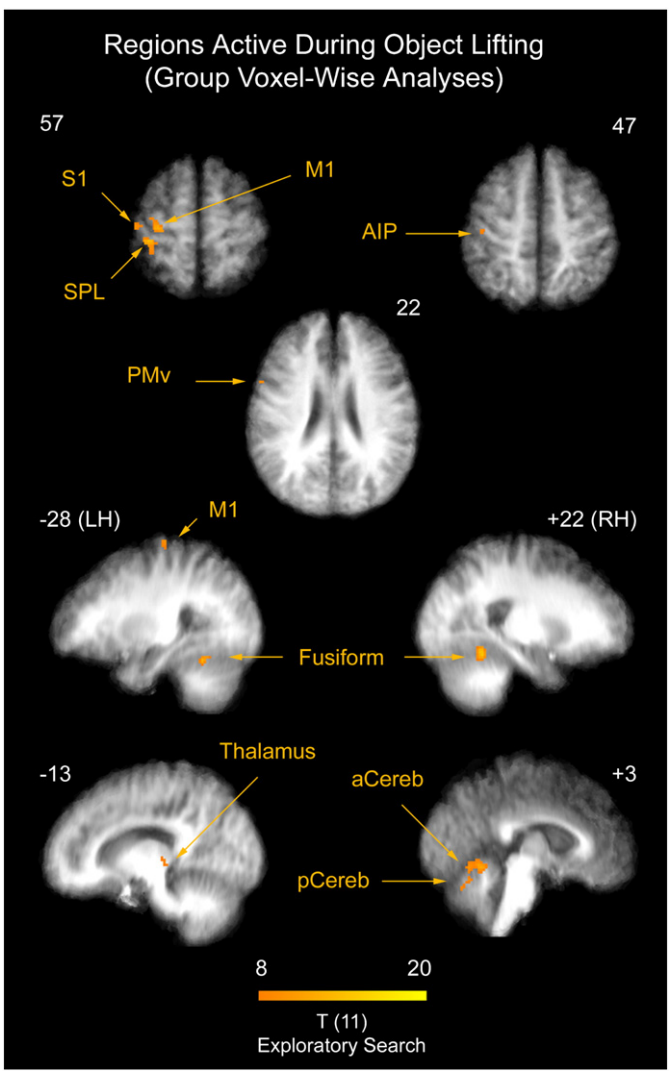

Fig. 4. Regions active during object lifting (group voxel-wise analyses). A group voxel-wise analysis identified areas involved in lifting objects (all conditions pooled > rest). The subtraction map was merged with the average anatomical MRI of all participants in standardized space and thresholded using the minimum given by Bonferroni correction and random-field theory for exploratory searches. See Table 2 for peak coordinates and *t*-statistical values.

Table 2
Regions active during object lifting (group voxel-wise analyses)

Brain Region			X	Y	Z	$T_{(11)}$
L	M1	Central sulcus, 'hand knob' area	−34	−22	56	12.9
L	S1	Postcentral gyrus	−50	−22	54	9.7
L	AIP	Postcentral/intraparietal sulci	−48	−20	48	8.7
L	SPL	Superior-parietal lobule	−42	−32	58	14.8
L	PMv	Precentral sulcus	−60	10	20	9.1
L	Thalamus		−14	−22	−2	8.4
L	Fus	Fusiform gyrus	−28	−52	−28	12.8
R	Fus	Fusiform gyrus	22	−50	−24	14.9
R	Cerebellum	Anterior lobule of cerebellum	4	−56	−18	12.8
		Posterior lobule of cerebellum	10	−70	−46	14.2

Coordinates are in MNI space.

fMRI experiment

In the DS–DW and SS–DW conditions, subjects always reported correctly that the heavier object was indeed heavier. In the SW illusion condition, however, subjects reported that the smaller of two objects of equal weight was heavier on average 89% of the time (11% SD; 63% to 100% range). This percentage was greater than chance ($P<0.001$), indicating that subjects experienced SW illusions.

Regions activated during object lifting

A group voxel-wise analysis used to identify areas activated during the lifting of objects revealed increased BOLD during object lifting (compared to rest) in the hand area of the left primary motor area (M1) in the central sulcus, the left primary somatosensory area (S1) in the postcentral gyrus, the left anterior intraparietal area (AIP) area in the postcentral sulcus close to where this sulcus joins the intraparietal sulcus, the anterior portion of the superior-parietal lobule (SPL), the left PMv in the precentral sulcus, the ventro-lateral portion of the left thalamus, the fusiform (Fus) gyrus in the two hemispheres, the right anterior lobule of the cerebellum above and anterior to the primary fissure of the cerebellum, and the right posterior lobule of the cerebellum below and posterior to the horizontal fissure of the

Table 3
Mean (and SD) coordinates for ROIs defined in each individual for subject-by-subject analyses

ROI		X	Y	Z
		Mean (SD)	Mean (SD)	Mean (SD)
L	M1	−36 (5)	−21 (5)	60 (7)
L	S1	−47 (3)	−24 (5)	60 (4)
L	AIP	−51 (4)	−26 (7)	52 (5)
L	SPL	−44 (7)	−38 (5)	60 (4)
L	PMv	−58 (4)	10 (6)	20 (6)
L	Thalamus	−15 (4)	−23 (4)	2 (4)
L	Fus	−28 (7)	−56 (8)	−23 (3)
R	Fus	29 (4)	−51 (7)	−23 (3)
R	aCereb	8 (6)	−64 (7)	−19 (8)
R	pCereb	14 (6)	−65 (5)	−50 (6)

Coordinates are in MNI space.

cerebellum (Fig. 4; Table 2). Based on these results, we formulated 'object-lifting' ROIs so that we could examine the more subtle effects of fMRI adaptation to the size, weight, or density of objects in key areas brain areas involved in lifting objects. We report in Table 3 the mean (and SD) coordinates for each 'object-lifting' ROI defined on an individual basis.

Group voxel-wise analyses

An exploratory search of the whole brain did not reveal any effects outside of the 'object-lifting' ROIs. The left S1, the left AIP, the left SPL, and the Fus in the two hemispheres responded more when subjects lifted objects with different sizes compared to objects with the same size (Fig. 5; Table 4) and the left M1 responded more when subjects lifted objects with different weights compared to objects with the same weight (Fig. 5; Table 4). The left PMv showed greater BOLD when subjects lifted objects of different densities compared to when they lifted objects with the same density (i.e. a size×weight interaction, which represents density, was significant; Fig. 6; Table 5). In contrast, the left M1, the left S1, and the left AIP responded less when subjects lifted objects with different densities compared to objects with the same density (Fig. 6; Table 5).

Our analyses also revealed that the left PMv was activated in the illusory contrast. In other words, the left PMv responded more when subjects falsely perceived differences in weight during the SW illusion versus the SS–SW condition (Fig. 6; Table 5). To determine whether or not this difference in BOLD was driven by differences in density or by differences in size (or both) between the two objects in the SW illusion, we conjoined the illusory contrast with the contrast identifying voxels that showed effects of density and conjoined the illusory contrast with the contrast identifying voxels that showed

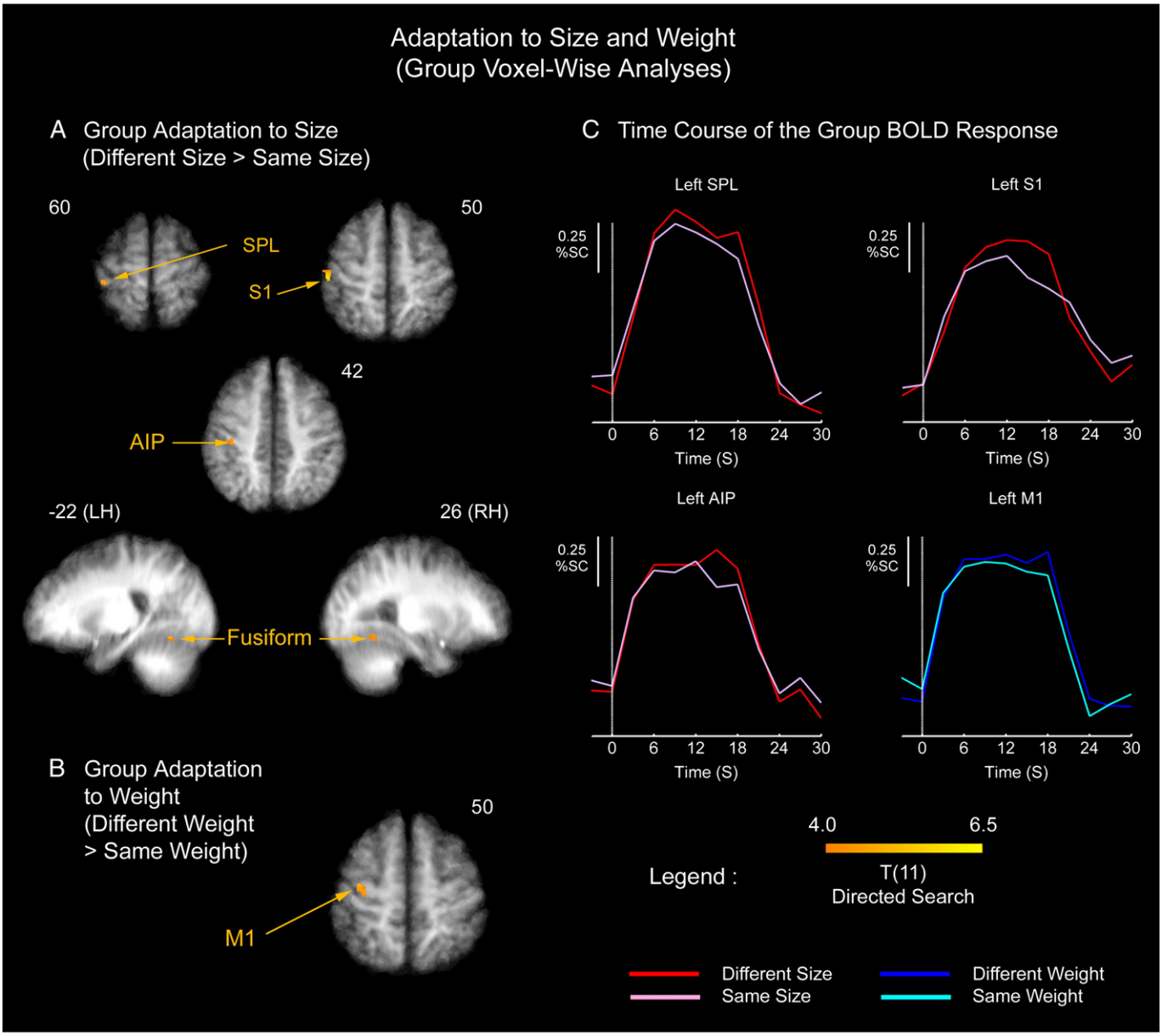

Fig. 5. Adaptation to size and weight (group voxel-wise analyses). Statistical maps, masked for all ROIs, were merged with the average anatomical MRI of all participants in standardized space and thresholded using the minimum given by Bonferroni correction and random-field theory for directed searches. (A) BOLD in the left SPL, the left S1, the left AIP, the left Fus, and the right Fus was greater when subjects lifted objects that differed in size compared to objects with the same size (adaptation to size). (B) BOLD in the left M1 was greater when subjects lifted objects that differed in weight compared to objects with the same weight (adaptation to weight). See Table 4 for peak coordinates and *t*-statistical values. (C) The graphs illustrate the time course of the percent-signal change (%SC) in the BOLD response of all subjects extracted from peak coordinates obtained in the group voxel-wise analyses. Red = effects of size (dark = different, light = same). Blue = effects of weight (dark = different, light = same).

Table 4
Adaptation to size and weight (group voxel-wise analyses)

			X	Y	Z	$T_{(11)}$
Different size > same size						
L	S1	Postcentral gyrus	−58	−24	48	6.4
L	AIP	Postcentral/intraparietal sulci	−40	−24	42	4.9
L	SPL	Superior-parietal lobule /postcentral gyrus	−42	−32	60	5.1
L	Fus	Fusiform gyrus	−22	−62	−20	5.1
R	Fus	Fusiform gyrus	26	−52	−22	5.5
Different weight > same weight						
L	M1	Central sulcus, 'hand knob' area	−38	−14	50	4.7

Coordinates are in MNI space.

Table 5
Effects of density and false perceptions of weight (group voxel-wise analyses)

			X	Y	Z	$T_{(11)}$
Different density > same density						
L	PMv	Precentral sulcus	−58	12	18	4.5
Same density > different density						
L	M1	Central sulcus, 'hand knob' area	−38	−12	50	5.6
L	S1	Postcentral gyrus	−34	−20	46	6.4
L	AIP	Superior-parietal lobule /postcentral gyrus	−38	−26	42	5.9
SW illusion > SS–SW						
L	PMv	Precentral sulcus	−60	8	22	4.3
SW illusion > SS–SW ∩ effect of density						
L	PMv	Precentral sulcus	−58	16	20	3.6

Coordinates are in MNI space.

effects of size. The first conjunction revealed that differences in density drove the increase in BOLD in the left PMv during the SW illusion (Fig. 6; Table 5). The second conjunction analysis, which conjoined the illusory contrast with the contrast identifying voxels that showed effects of size, did not reveal any significant changes in BOLD that were common to both contrasts.

In summary, the left S1, the left AIP, the left SPL, and the Fus in the two hemispheres adapted to object size whereas the left M1 adapted to object weight. The left PMv showed adaptation when objects of the same density were lifted, whereas the left M1, the left S1, and the left AIP responded more when objects of the same density were lifted. Conjunction analyses revealed that changes in the left PMv in the SW illusion were driven by differences in the density between two objects

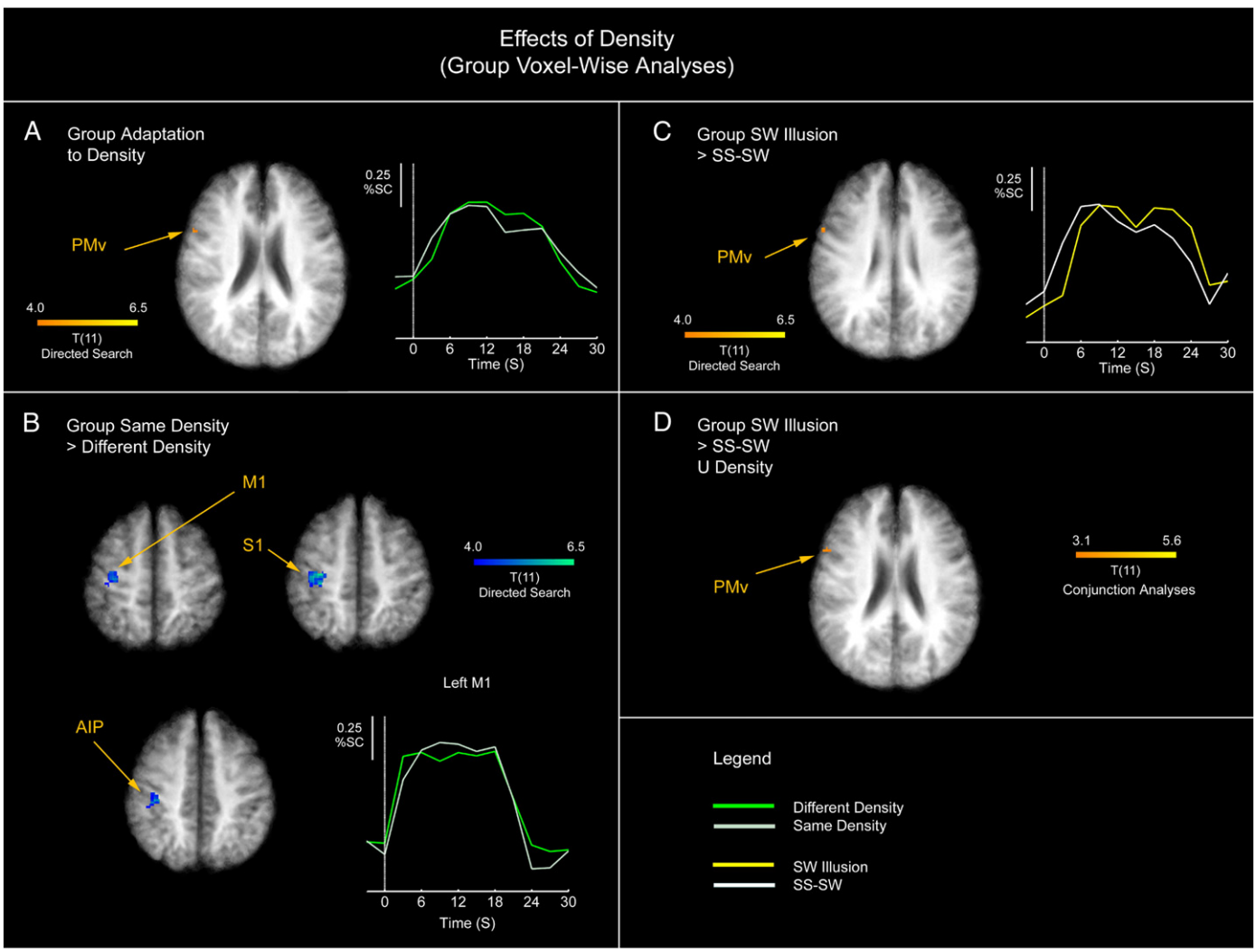

Fig. 6. Effects of density (group voxel-wise analyses). Statistical maps, masked for all ROIs, were merged with the average anatomical MRI of all participants in standardized space. Also, we present graphs that illustrate the average time course of the percent-signal change (%SC) in the BOLD response of all subjects extracted from peak coordinates obtained in the group voxel-wise analyses. Green = effects of density (dark = different, light = same). Yellow = SW illusion. White = SS–SW. (A) BOLD in the left PMv was greater when subjects lifted objects that differed in density compared to objects with the same density (adaptation to density). (B) BOLD in the left M1, the left S1, and the left AIP was greater when subjects lifted objects that had the same density compared to objects that differed in density. (C) BOLD in the left PMv was greater during the SW Illusion compared to the SS–SW condition. (D) Conjunction analyses revealed that changes in BOLD in the left PMv during SW illusions were driven by differences in the density of the two objects in the SW illusion condition ($P_{uncorr} < 0.005$ in both contrasts that were conjoined). See Table 5 for peak coordinates and *t*-statistical values.

Table 6
Effects of size, weight, and density (subject-by-subject analyses)

		Direction of effect	$F_{(1,11)}$
Effect of size			
L	S1	Different > same	5.50
L	AIP	Different > same	11.98
L	SPL	Different > same	12.07
L	Fus	Different > same	7.11
R	Fus	Different > Same	14.08
Effect of weight			
L	M1	Different > same	7.02
Effect of density (size × weight)			
L	PMv	Different > same	11.86
L	S1	Same > different	20.96
L	AIP	Same > different	5.26
L	M1	Same > different	9.69

of different sizes that weighed the same and not by differences in their physical size.

Subject-by-subject analyses

ANOVA revealed greater BOLD in the left S1, the left AIP, the left SPL, and the Fus in the two hemispheres when subjects lifted objects with different sizes compared to objects with the same size (Table 6, all $P<0.05$) and did not reveal any significant main effects of size in any of the other 'object-lifting' ROIs. ANOVA revealed greater BOLD in the left M1 when subjects lifted objects with different weights compared to objects with the same weight (Table 6, $P<0.05$) and did not reveal any main effects of weight in any of the other 'object-lifting' ROIs. ANOVA also revealed that BOLD increased with changes in density in the left PMv and decreased with changes in density in the left M1, the left S1, and the left AIP (Table 6, all size × weight interactions were significant, $P<0.05$). The mean (and confidence intervals for within-subject contrasts) percent-signal change in BOLD obtained in each 'object-lifting' ROI are plotted for each condition in Fig. 7. Pair-wise comparisons revealed greater BOLD in the left PMv during the SW illusion compared to the SS–SW condition (Fig. 7; $P<0.05$).

In summary, results from the subject-by-subject analyses coincided well with results from the group voxel-wise analyses. The left S1, the left AIP, the left SPL, and the Fus in the two hemispheres adapted to size whereas the left M1 adapted to weight. The left PMv showed adaptation when objects of the same density were lifted, whereas the left M1, the left S1, and the left AIP responded more when objects of the same density were lifted.

Discussion

As we show in our behavioral experiment and as others have shown (Flanagan and Beltzner, 2000; Grandy and Westwood, 2006; Chang et al., 2008), people quickly learn to scale their fingertip forces to the actual weight of objects during SW illusions, even though they fail to learn perceptually that both objects have in fact the same weight. The question then arises as to what the separate neural mechanisms are for determining the perceived heaviness of objects and the predicted weight of these same objects during lifting. Our imaging data reveal that sensory areas in the parietal and temporal cortex adapted to the size of objects whereas the primary motor area (M1) contralateral to the lifting hand adapted to the weight of objects.

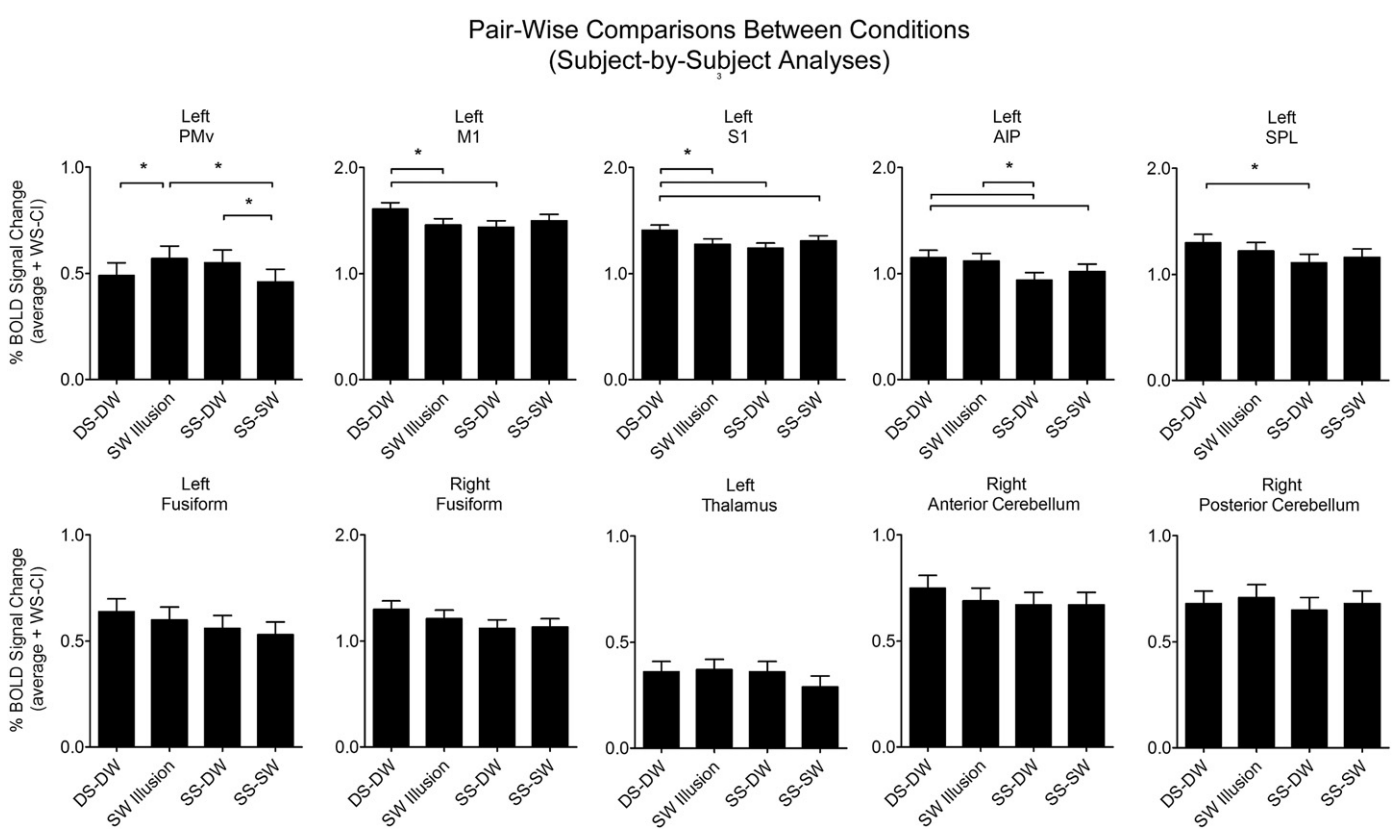

Fig. 7. Pair-wise comparisons between conditions (subject-by-subject analyses). In addition to testing for effects of size, weight, and density (i.e. size × weight interactions), we tested also for differences between conditions on BOLD extracted in ROIs that were defined on a subject-by-subject basis using Tukey's HSD pair-wise comparison tests, which corrected for multiple comparisons. Graphs show the mean percent-signal change in BOLD extracted from ROIs that were defined on a subject-by-subject basis + 95% confidence intervals for within-subject contrasts (Loftus and Masson, 1994). *Y*-axes are scaled differently to better distinguish differences between conditions. Asterisks (*) denote significant differences ($P<0.05$).

The left ventral premotor area (PMv), which did not adapt to either the size or the weight of objects, adapted instead to the density of objects, and responded more when subjects falsely perceived differences in weight between objects during the SW illusion. Taken together, we conclude that the real-world properties of objects, such as size and weight, are computed by sensory areas in the parietal and temporal cortices and by M1 respectively, whereas the perceived heaviness of objects, presumably based on their apparent density, is computed by PMv, a higher-order area well placed to integrate sensory information about the size of objects and the weight of those objects after they have been lifted.

Methodological issues

We used a block design, which provided more statistical power than an event-related design would have done, given the number of conditions that were required to answer the questions we had set. One important limitation of a block design, however, is that it does not allow one to compare BOLD between trials. It might have been interesting to compare BOLD during initial trials when people were unable to scale forces for weight to BOLD in subsequent trials when people were able to scale forces for weight. Although this is one approach for identifying brain areas responsible for the updating of sensorimotor predictions, it does not by itself identify the brain regions that generate SW illusions. Therefore, we felt it was important to use a block design that included additional conditions that would permit us to measure adaptation to the size, the weight, or the density of objects independently from one another, and to determine which of these features drove SW illusions. Had we used an event-related design by temporally spacing each lifting trial further apart, the fMRI session would have been far too long in duration to include all the conditions that we wished to examine. Note that our block design did still allow us to measure adaptation to weight (Figs. 1A, B; see Overview of fMRI experiment in Methods), and this consequently allowed us to infer which brain areas are involved in the updating of sensorimotor predictions.

Although an event-related design could have also enabled us to decouple artifacts caused by movements from task-related activations (Culham et al., 2003), we decided instead to minimize head movements. Subjects underwent a training session prior to scanning so that they could practice lifting objects with minimal motion in the shoulder. An arm brace was used during both the training and scanning sessions to restrict shoulder movement and the conduction of arm movements to the head (Fig. 1D). Moreover, inspection of the fMRI data revealed that head movements associated with lifting objects, when present, were within an acceptable range of less than 1 mm (see Supplementary material).

Given the relationship between the size, the weight, and the density of objects, it was not possible to balance all these features across all conditions (see Table 1). We believe, however, that our contrasts did still allow us to measure adaptation that was unique to each of these features. What is most crucial in an fMRI adaptation study is the comparison of the BOLD response when a feature changes across trials to the BOLD response when that same feature is maintained across all trials while at the same time removing any effects that may be driven by other features. Because we used a block design in which subjects lifted each object only three times, one might argue that our results could have perhaps been influenced by differences in the actual weight of objects across conditions. But expectations of weight, and the subsequent scaling of forces applied during lifting, depend first and foremost on any associations of weight to a particular visual appearance of an object, which may or may not be the same as its actual weight. In fact, in the fMRI experiment, there was a high a degree of uncertainty at the beginning of each block as to what a particular object might weigh. This is because size did not always reflect the weight of objects (see Table 1).

Finally, although the precise neural events that account for fMRI adaptation are not yet understood, what is clear is that repetition suppression effects occur at the level of both cortical neurons and local hemodynamic changes (for review, see Grill-Spector et al., 2006). Therefore, one can make inferences as to whether or not a given brain area codes for a particular feature by contrasting BOLD when this feature changes across trials with BOLD when this feature is maintained across trials. As a consequence, fMRI adaptation has proven to be a very useful technique for asking what is and what is not processed by a brain area.

Adaptation to size

The left AIP, the left SPL, and the Fus in the two hemispheres adapted to object size (Fig. 5). Adaptation to the visual presentation of objects in the Fus, however, has been shown to be invariant to size in earlier fMRI studies (Grill-Spector et al., 1999; Vuilleumier et al., 2002). In our experiment, changes in size were confounded with shape, because the objects had been designed to be lifted comfortably in the scanner. Big objects were wider but not taller than the small objects so that subjects would have enough room inside the scanner to lift big objects. Thus, Fus could have been adapting to shape rather than size (Kourtzi and Kanwisher, 2001). Alternatively, ventral-stream areas, such as Fus, might process information regarding object size depending on task demands (Cavina-Pratesi et al., 2007). Given that subjects assessed the heaviness of objects in which size usually predicts weight, it is plausible that top-down influences may have caused Fus to process size information. Further investigation, however, will be required to determine whether or not this is in fact the case. Yet, it certainly seems likely that ventral-stream areas would be recruited when initial grip and load forces are calibrated, given that size (together with a range of visual cues) can provide information about the expected weight of an object (Milner and Goodale, 2006).

Adaptation to object size in sensorimotor areas could reflect the transformation of visual information about the geometry of the objects into the motor commands that specify movements required to pick up the objects. Gardner et al. (2007a,b) found similarities in activity in neurons in the macaque monkey recorded from S1, AIP and Brodmann area 5 in the SPL in which firing was greatest during the approach, contact, and grasping stages of object lifting. This contrasted with activity in M1, in which different populations of M1 neurons each fired at different stages of object lifting. This might explain why S1, AIP and SPL adapted to object size whereas M1 adapted to object weight. In addition, it is possible that adaptation to object size in somatosensory areas may have also reflected the processing of proprioceptive feedback from rotational forces (i.e. torque) applied during lifting. Such input could conceivably provide volumetric information about the size of objects. Note that subjects manipulated all objects using a similar grasping configuration applied to identical wooden handles attached to the top of each object (Figs. 1A, D). Therefore, information about size could not have been obtained haptically by grasping the objects.

Adaptation to weight

The left M1 adapted to object weight (Fig. 5). As it turns out, electrophysiological recordings performed in macaque monkeys reveal that different populations of M1 neurons code for the specific forces required to lift objects of different weights (Picard and Smith, 1992). M1 lesions result in a reduction in hand strength and impairments in generating independent finger movements, whereas lesions to somatosensory areas include the loss of finger coordination and lead to the adoption of the general strategy of applying excessive forces during lifting (Brochier et al., 1999). In a histological and retrograde tracing study of corticospinal neurons, Dum and Strick (1991) classified nearly one third of corticospinal neurons that arise

from M1 as being 'large' with a diameter of greater than 19 μm, and found that nearly eighty percent of all corticospinal neurons in the frontal lobes that were classified as being 'large' arose from M1. Although Dum and Strick (1991) also showed that a number of smaller corticospinal neurons do arise from premotor areas, it is the larger corticospinal neurons from M1 that are thought to be important for generating forces because only they exert a direct influence on motor neurons in the spinal cord (Lemon et al., 2002).

There is also evidence that M1 uses feedback from previous experiences to update sensorimotor predictions for lifting objects, presumably by updating internal representations concerning the weight of objects. In a transcranial magnetic stimulation (TMS) study, Chouinard et al. (2005) showed that M1 can form memory traces related to forces applied during lifting. When subjects lifted an object without TMS being applied, they generated fingertip forces that were based on the remembered weight of that object on the previous trial. When TMS was applied over M1, however, subjects could no longer scale forces on the basis of the weight of previously lifted objects, and instead generated forces until a sufficient vertical force was reached to overcome gravity, without taking into consideration their previous experience with that object. Given that fMRI adaptation has sometimes been associated with the formation of certain forms of memories (Henson, 2003), the observed fMRI adaptation in M1 could perhaps reflect the updating of sensorimotor predictions for lifting objects in subsequent attempts.

Lack of effects in cerebellum

Although one cannot establish the absence of an effect with classical inference (i.e., accept the null hypothesis), it was surprising to us that we could not find any effects of size, weight, or density in the cerebellum nor any differences in BOLD in the SW illusion versus the SS–SW condition. It might be the case that the cerebellum is not concerned with the computation of real-world properties of objects, such as size and weight, or the apparent densities of objects. After all, the most common deficit associated with cerebellar damage is ataxia, a neurological disorder characterized by the loss in ability to co-ordinate movements during execution (Fine et al. 2002). Even though there is increasing evidence to support the idea that the cerebellum might also fulfill a number of higher-order functions (Gordon, 2007), its role in SW illusions remains unclear. To our knowledge, there is no direct evidence from the patient literature to show that cerebellar damage reduces SW illusions. On the contrary, Rabe et al. (2007) reveals that SW illusions continue to persist in patients with cerebellar degeneration and is not reduced compared to neurologically-intact control subjects.

Effects of density and false perceptions of weight

The left M1, the left S1, and the left AIP responded more when objects of the same density were lifted (Fig. 6). This finding may be directly related to the observed release from adaptation in these same regions when size or weight was varied. In the fMRI experiment, when both size and weight were varied, density was purposely held constant. This meant that an area interested in the physical properties of objects such as size and weight could show the greatest BOLD response in the DS–DW condition in which both the size and weight of two objects were different—a condition in which the two objects had the same density. Our results reveal that this was indeed the case for the left M1, S1, and AIP (Fig. 7). Thus, it would appear that the effects of size and of weight summated in these three areas.

In contrast, the left PMv adapted to object density (Fig. 6). It is conceivable that density, which requires the integration of sensory information about the size of objects and the weight of objects once they are lifted, needs to be processed by a higher-order area such as PMv. There is a good deal of evidence that PMv plays a central role in a wide range of cognitive functions related to actions (Rizzolatti et al. 2002) and is well placed to integrate different forms of sensory experiences through its connections with other brain structures (Barbas and Pandya, 1987; Rizzolatti et al., 1988). But even though PMv adapts to density, the question remains as to why this area would also play a role in the perception of the heaviness of objects (Fig. 6).

One possible explanation is that density is a more salient feature for making cognitive judgments about the heaviness of an object than its actual mass (Grandy and Westwood 2006). Evidence for this notion comes from behavioral studies that demonstrate material–weight illusions, in which the perceived heaviness of equal-weight objects depends of the perception of their material properties, and thus their perceived density (Ross, 1969; Harshfield and DeHardt, 1970). There is also neuropsychological evidence to support that PMv is critical for generating SW illusions. Halstead (1945) describes a patient with a resection in the left inferior-frontal cortex, which appeared to encompass PMv, who was not fooled by the SW illusion. This patient was able to make accurate weight discriminations for equally weighted objects that varied in size while neurologically-intact control subjects could not. Together, this evidence supports our claim that weight perception is most likely fulfilled by PMv, a higher-order area well placed to integrate sensory information about the size of objects and the weight of those objects after they have been lifted.

Of interest, earlier fMRI studies have shown that that PMv is involved in judging the heaviness of objects when people also observe videos of either themselves or others lift objects (Grezes et al., 2004; Hamilton et al., 2006). It has been suggested that PMv might mediate the understanding of actions (Rizzolatti and Luppino 2001). In the macaque monkey, PMv contains a class of neurons called *mirror neurons* (Gallese et al., 1996), which discharge both when monkeys make a particular action and when they observe another monkey or human make the same action. Functional neuroimaging has shown evidence that suggest that this same class of neurons might also exist in PMv in humans (Grezes et al., 2003; Chouinard and Paus, 2006). It could be the case that similar types of processing take place in PMv when people make weight judgments about objects that they lift and about objects that they see others lift.

Concluding remarks

It would be inefficient, and perhaps even maladaptive, to persist in lifting objects using forces that did not reflect their actual weight. Of course, the first time one lifts a particular object, the only information one has about the expected weight of that object comes from its appearance and any learned associations with that appearance. But as we demonstrated in our behavioral experiment, after lifting the object a few times, the sensorimotor system learns to quickly calibrate fingertip forces to its actual weight, presumably on the basis of previous encounters with that object. M1, which adapted to the weight of the objects in our fMRI experiment, may provide a substrate for updating internal representations of the actual weight of objects. PMv adapted instead to the density of objects and responded more when subjects falsely perceived differences in weight between objects in the SW illusion compared to when they lifted objects that were identical in size and weight. This finding is consistent with the notion that density, which requires the integration of sensory information about the size and weight of objects, is a more salient feature for cognitive judgments of heaviness than the actual weight of objects.

Acknowledgments

We thank Mrs. Joy Williams and Dr. Haitao Yang for providing us with technical support. This work was supported by the Canadian Institutes of Health Research.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2008.08.023.

References

Barbas, H., Pandya, D.N., 1987. Architecture and frontal cortical connections of the premotor cortex (area 6) in the rhesus monkey. J. Comp. Neurol. 256, 211–228.

Brett, M., Johnsrude, I.S., Owen, A.M., 2002. The problem of functional localization in the human brain. Nat. Rev. Neurosci. 3, 243–249.

Brochier, T., Boudreau, M.J., Pare, M., Smith, A.M., 1999. The effects of muscimol inactivation of small regions of motor and somatosensory cortex on independent finger movements and force control in the precision grip. Exp. Brain Res. 128, 31–40.

Cavina-Pratesi, C., Goodale, M.A., Culham, J.C., 2007. FMRI reveals a dissociation between grasping and perceiving the size of real 3D objects. PLoS ONE 2, e424.

Chang, E.C., Flanagan, J.R., Goodale, M.A., 2008. The intermanual transfer of anticipatory force control in precision grip lifting is not influenced by the perception of weight. Exp. Brain Res. 185, 319–329.

Charpentier, A., 1891. Analyse expérimentale: de quelques éléments de la sensation de poids. Arch. Physiol. Norm. Pathol. 3, 122–135.

Chouinard, P.A., Paus, T., 2006. The primary motor and premotor areas of the human cerebral cortex. Neuroscientist 12, 143–152.

Chouinard, P.A., Leonard, G., Paus, T., 2005. Role of the primary motor and dorsal premotor cortices in the anticipation of forces during object lifting. J. Neurosci. 25, 2277–2284.

Collins, D.L., Neelin, P., Peters, T.M., Evans, A.C., 1994. Automatic 3D intersubject registration of MR volumetric data in standardized Talairach space. J. Comput. Assist. Tomogr. 18, 192–205.

Cox, R.W., 1996. AFNI: software for analysis and visualization of functional magnetic resonance neuroimages. Comput. Biomed. Res. 29, 162–173.

Culham, J.C., Danckert, S.L., DeSouza, J.F., Gati, J.S., Menon, R.S., Goodale, M.A., 2003. Visually guided grasping produces fMRI activation in dorsal but not ventral stream brain areas. Exp. Brain Res. 153, 180–189.

Dinstein, I., Hasson, U., Rubin, N., Heeger, D.J., 2007. Brain areas selective for both observed and executed movements. J. Neurophysiol. 98, 1415–1427.

Dum, R.P., Strick, P.L., 1991. The origin of corticospinal projections from the premotor areas in the frontal lobe. J. Neurosci. 11, 667–689.

Fine, E.J., Ionita, C.C., Lohr, L., 2002. The history of the development of the cerebellar examination. Semin. Neurol. 22, 375–384.

Flanagan, J.R., Beltzner, M.A., 2000. Independence of perceptual and sensorimotor predictions in the size–weight illusion. Nat. Neurosci. 3, 737–741.

Frey, S.H., Vinton, D., Norlund, R., Grafton, S.T., 2005. Cortical topography of human anterior intraparietal cortex active during visually guided grasping. Brain Res. Cogn. Brain Res. 23, 397–405.

Gallese, V., Fadiga, L., Fogassi, L., Rizzolatti, G., 1996. Action recognition in the premotor cortex. Brain 119 (Pt 2), 593–609.

Gardner, E.P., Babu, K.S., Reitzen, S.D., Ghosh, S., Brown, A.S., Chen, J., Hall, A.L., Herzlinger, M.D., Kohlenstein, J.B., Ro, J.Y., 2007a. Neurophysiology of prehension. I. Posterior parietal cortex and object-oriented hand behaviors. J. Neurophysiol. 97, 387–406.

Gardner, E.P., Ro, J.Y., Babu, K.S., Ghosh, S., 2007b. Neurophysiology of prehension. II. Response diversity in primary somatosensory (S-I) and motor (M-I) cortices. J. Neurophysiol. 97, 1656–1670.

Glover, G.H., 1999. Deconvolution of impulse response in event-related BOLD fMRI. Neuroimage 9, 416–429.

Gordon, N., 2007. The cerebellum and cognition. Eur. J. Paediatr. Neurol. 11, 232–234.

Grandy, M.S., Westwood, D.A., 2006. Opposite perceptual and sensorimotor responses to a size–weight illusion. J. Neurophysiol. 95, 3887–3892.

Grezes, J., Armony, J.L., Rowe, J., Passingham, R.E., 2003. Activations related to “mirror” and “canonical” neurones in the human brain: an fMRI study. Neuroimage 18, 928–937.

Grezes, J., Frith, C.D., Passingham, R.E., 2004. Inferring false beliefs from the actions of oneself and others: an fMRI study. Neuroimage 21, 744–750.

Grill-Spector, K., Kushnir, T., Edelman, S., Avidan, G., Itzchak, Y., Malach, R., 1999. Differential processing of objects under various viewing conditions in the human lateral occipital complex. Neuron 24, 187–203.

Grill-Spector, K., Henson, R., Martin, A., 2006. Repetition and the brain: neural models of stimulus-specific effects. Trends Cogn. Sci. 10, 14–23.

Halstead, W.C., 1945. Brain injuries and the higher levels of consciousness. Trauma of the Central Nervous System. Res. Publ. Assoc. Res. N., 24. Williams and Wilkins, Baltimore, pp. 480–506.

Hamilton, A.F., Wolpert, D.M., Frith, U., Grafton, S.T., 2006. Where does your own action influence your perception of another person's action in the brain? Neuroimage 29, 524–535.

Harshfield, S.P., DeHardt, D.C., 1970. Weight judgment as a function of apparent density of objects. Psychonom. Sci. 20, 365–366.

Henson, R.N., 2003. Neuroimaging studies of priming. Prog. Neurobiol. 70, 53–81.

Johansson, R.S., Westling, G., 1988. Coordinated isometric muscle commands adequately and erroneously programmed for the weight during lifting task with precision grip. Exp. Brain Res. 71, 59–71.

Kawai, S., 2002. Heaviness perception. II. Contributions of object weight, haptic size, and density to the accurate perception of heaviness or lightness. Exp. Brain Res. 147, 23–28.

Kourtzi, Z., Kanwisher, N., 2001. Representation of perceived object shape by the human lateral occipital complex. Science 293, 1506–1509.

Lemon, R.N., Maier, M.A., Armand, J., Kirkwood, P.A., Yang, H.W., 2002. Functional differences in corticospinal projections from macaque primary motor cortex and supplementary motor area. Adv. Exp. Med. Biol. 508, 425–434.

Loftus, G.R., Masson, M.E.J., 1994. Using confidence intervals in within-subjects designs. Psychonom. Bull. Rev. 1, 476–490.

Milner, A.D., Goodale, M.A., 2006. The Visual Brain in Action, second ed. Oxford University Press, Oxford.

Nichols, T., Brett, M., Andersson, J., Wager, T., Poline, J.B., 2005. Valid conjunction inference with the minimum statistic. Neuroimage 25, 653–660.

Ogawa, S., Tank, D.W., Menon, R., Ellermann, J.M., Kim, S.G., Merkle, H., Ugurbil, K., 1992. Intrinsic signal changes accompanying sensory stimulation: functional brain mapping with magnetic resonance imaging. Proc. Natl. Acad. Sci. U. S. A. 89, 5951–5955.

Oldfield, R.C., 1971. The assessment and analysis of handedness: the Edinburgh inventory. Neuropsychologia 9, 97–113.

Picard, N., Smith, A.M., 1992. Primary motor cortical activity related to the weight and texture of grasped objects in the monkey. J. Neurophysiol. 68, 1867–1881.

Price, C.J., Friston, K.J., 1997. Cognitive conjunction: a new approach to brain activation experiments. Neuroimage 5, 261–270.

Rabe, K., Brandauer, B., Li, Y., Timmann, D., Hermsdörfer, J., 2007. Size–weight illusion and grip force anticipation in patients with cerebellar degeneration. 37th Annual Meeting of the Society for Neuroscience. San Diego, CA, USA.

Rizzolatti, G., Luppino, G., 2001. The cortical motor system. Neuron 31, 889–901.

Rizzolatti, G., Camarda, R., Fogassi, L., Gentilucci, M., Luppino, G., Matelli, M., 1988. Functional organization of inferior area 6 in the macaque monkey. II. Area F5 and the control of distal movements. Exp. Brain Res. 71, 491–507.

Rizzolatti, G., Fogassi, L., Gallese, V., 2002. Motor and cognitive functions of the ventral premotor cortex. Curr. Opin. Neurobiol. 12, 149–154.

Ross, H.E., 1969. When is a weight not illusory? Q. J. Exp. Psychol. 21, 346–355.

Vuilleumier, P., Henson, R.N., Driver, J., Dolan, R.J., 2002. Multiple levels of visual object constancy revealed by event-related fMRI of repetition priming. Nat. Neurosci. 5, 491–499.

Worsley, K.J., Evans, A.C., Marrett, S., Neelin, P., 1992. Determining the number of statistically significant areas of activation in subtracted activation studies from PET. J. Cereb. Blood Flow Metab. 12, 900–918.

Worsley, K.J., Marrett, S., Neelin, P., Vandal, A.C., Friston, K.J., Evans, A.C., 1996. A unified statistical approach for determining significant signals in images of cerebral activation. Hum. Brain Mapp. 4, 58–73.

Worsley, K.J., Liao, C.H., Aston, J., Petre, V., Duncan, G.H., Morales, F., Evans, A.C., 2002. A general statistical analysis for fMRI data. Neuroimage 15, 1–15.

Yousry, T.A., Schmid, U.D., Alkadhi, H., Schmidt, D., Peraud, A., Buettner, A., Winkler, P., 1997. Localization of the motor hand area to a knob on the precentral gyrus. A new landmark. Brain 120, 141–157.

Zwislocki, J.J., Goodman, D.A., 1980. Absolute scaling of sensory magnitudes: a validation. Percept. Psychophys. 28, 28–38.